DE PAUL INTERNATIONAL RESIDENTIAL SCHOOL, MYSURU

| Managed by Vincentian Fathers |

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Introduction

H4H (Humanoid for Humanity) is an AI-driven social innovation initiative that uses humanoid robotics, artificial intelligence, and smart technologies to protect human life, improve mental well-being, and create safer communities. The core philosophy of H4H is that technology should serve humanity, especially in areas where safety, mental health, and social protection are critical.

With increasing academic pressure, stress, and emotional challenges among students, H4H provides an AI-enabled humanoid system that acts as a safe, non-judgmental companion. The system helps students express their emotions freely, offers basic mental-wellness guidance, detects stress patterns, and encourages early intervention by guiding students toward professional support when required. This approach helps reduce anxiety, emotional isolation, and mental health stigma.

H4H also focuses on safeguarding vulnerable groups such as children and women through AI-based protection and tracking systems. The solution includes real-time location tracking, emergency alert mechanisms, and distress detection using behavior and movement analysis. In critical situations, the system automatically notifies parents, guardians, and authorities, enabling rapid response and preventive action. H4H functions as a silent digital protector that prioritizes safety and dignity.

To enhance public safety and reduce road accidents, H4H integrates AI-powered traffic management solutions. These systems optimize traffic signals, identify accident-prone zones, monitor vehicles and pedestrians in real time, and generate instant alerts during accidents. Predictive analysis helps prevent collisions before they occur, significantly improving road safety and saving human lives.

To make this technology interactive and accessible for students, educators, and young innovators, H4H is developed using PictoBlox, an AI and robotics-based coding platform by STEMpedia. PictoBlox supports both block-based coding and Python, enabling easy integration of artificial intelligence, IoT, computer vision, and automation into real-world applications. This empowers learners to transform innovative ideas into practical, socially impactful solutions.

Team Introduction

Our project, **"H4H – Humanoid for Humanity," developed using PictoBlox**, was created by a team of three passionate and innovative students committed to addressing real-world human and societal challenges through technology. Each team member contributed unique strengths and ideas, enabling us to collaboratively design and develop an intelligent, AI-driven solution focused on mental health support, human safety, and life protection. Through teamwork, creativity, and problem-solving, we transformed innovative concepts into a practical and socially impactful H4H system..

***Mishika S* – Hardware & Circuit Design Lead**

She designed and set up the **H4H system**, working with safety and monitoring sensors and integrating them with the **Quarkey/Arduino control board**. She ensured proper wiring and smooth hardware–AI interaction for reliable system performance.

***Namit Mohan* – Programming & Automation Specialist**

He led the software development of the H4H system using PictoBlox. Her work included developing block-based and Python programs to control sensors, automate safety responses, and manage system operations. She also integrated AI models for mental health pattern detection, safety monitoring, and IoT-based dashboard visualization to enable real-time alerts and intelligent decision-making.

***Suhani Somashekar* – Research, Documentation & Presentation**

She conducted research on **H4H technologies**, AI-based safety systems, and intelligent system design.
She was responsible for project documentation, workflow management, and preparing the final presentation for exhibition and judging.

Theme of the Project:

The theme of our project is "Technology for Humanity and Human Safety." This theme highlights the use of modern tools and intelligent technologies to address critical human and societal challenges. By integrating artificial intelligence (AI), automation, and the Internet of Things (IoT), the H4H project aims to deliver innovative, practical, and scalable solutions focused on mental health support, human safety, abuse prevention, and accident reduction. The system is developed using PictoBlox, a STEM-based coding platform that enables the creation of interactive, AI-powered, and socially impactful solutions.

Reason for Selecting the Project:

Human safety, mental well-being, and social protection have become critical global challenges affecting people of all ages, especially students, women, and children. Increasing stress levels, safety risks, road accidents, and delayed emergency response systems demand immediate attention and intelligent, human-centered solutions that are affordable, reliable, and easy to deploy.

The H4H (Humanoid for Humanity) system offers a promising approach by using artificial intelligence, humanoid interaction, automation, and the Internet of Things (IoT) to support mental health, enhance personal safety, and reduce accidents. We chose this project for the following reasons:

• To develop an AI-driven system focused on human well-being and life protection.
• To utilize PictoBlox for learning and implementing sensor integration, coding, AI, and automation.
• To create a smart, AI-powered, and IoT-enabled solution for real-time monitoring and alerts.
• To design a scalable model that can be deployed in schools, homes, public spaces, and smart cities.
• To contribute to the betterment of society by promoting safety, mental wellness, and responsible use of technology.

Alignment with United Nations Sustainable Development Goals (SDGs)

Our project supports and aligns with multiple goals set by the United Nations 2030 Agenda for Sustainable Development:

Goal 3: Good Health and Well-Being

• Supports mental health and emotional well-being of students through AI-enabled humanoid interaction.
• Helps in early stress detection, emotional support, and guidance for seeking professional help.
• Promotes overall well-being by using technology as a safe and supportive companion.

Goal 10: Reduced Inequalities

• Provides accessible and affordable AI-based safety and mental health solutions for all sections of society.
• Ensures support for students, women, and vulnerable groups, regardless of social or economic background.
• Bridges the gap in access to safety and well-being technologies using low-cost platforms like PictoBlox.

Goal 16: Peace, Justice and Strong Institutions

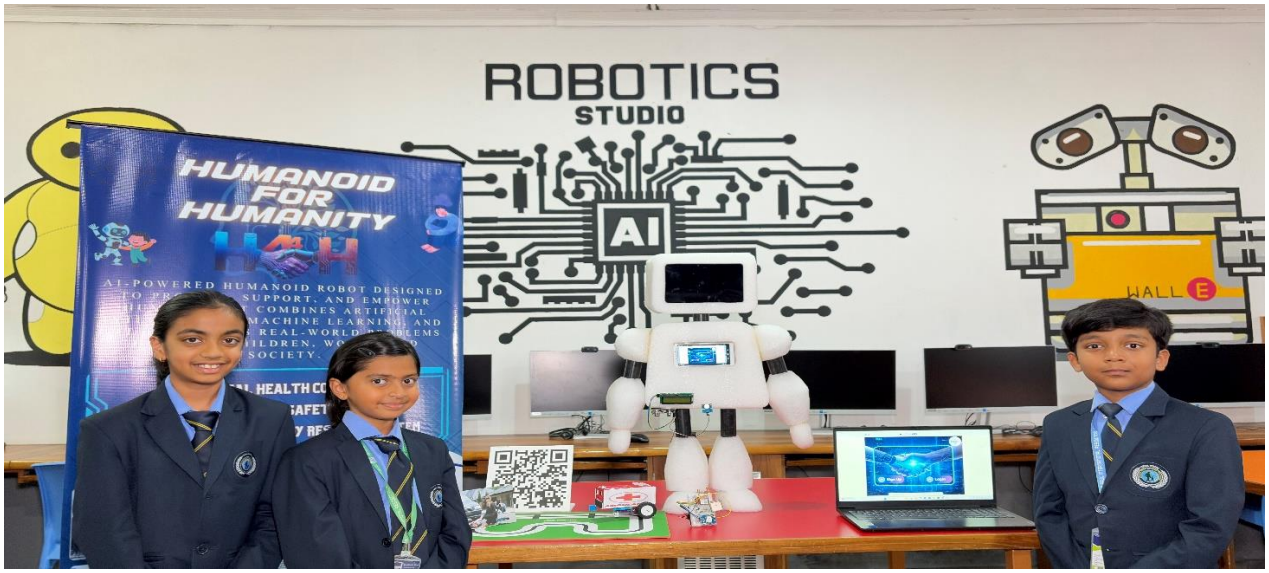
• Enhances human safety and protection, especially for women and children, using AI-based alert and tracking systems.
• Supports crime prevention, emergency response, and accident reduction through intelligent monitoring.
• Strengthens trust in technology by promoting safe, ethical, and responsible AI usage for public safety.

Overview of the System:

The H4H (Humanoid for Humanity) system is a smart, AI-driven solution designed to support human well-being, safety, and life protection through intelligent interaction and automation. The system is designed and controlled using PictoBlox, a graphical programming platform that enables easy integration of sensors, actuators, AI models, and IoT functionalities. H4H aims to provide an automated and intelligent solution for mental health support, human safety, abuse prevention, and accident reduction, especially in environments where immediate assistance and monitoring are critical.

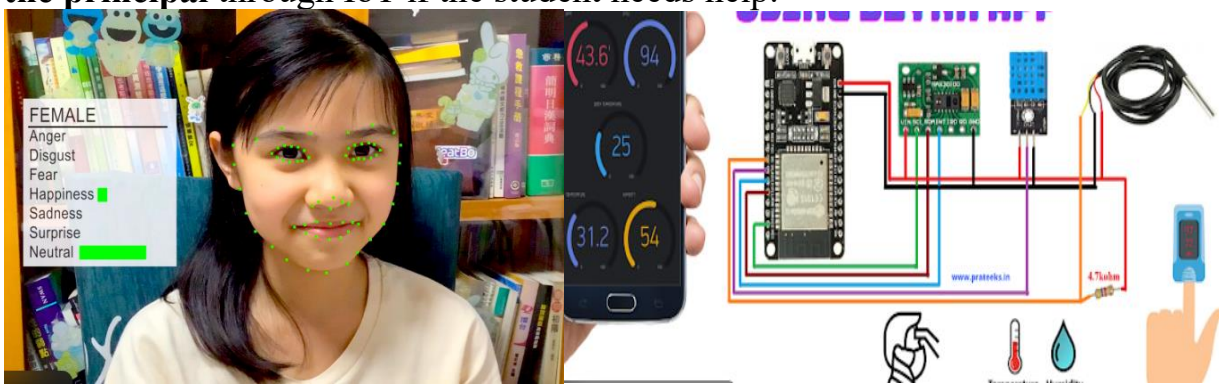
Working Principle:

The basic working principle of the H4H system is **AI-based sensing, analysis, and response**. The system continuously collects data from various sensors related to human behavior, environment, and safety conditions. This data is processed using AI models to detect stress patterns, unsafe situations, or emergency conditions. Based on the analysis, the system automatically triggers appropriate responses such as alerts, guidance, or notifications through IoT-based communication, ensuring timely support and protection of human life.



Mental Health AI Companion Robot

A **Mental Health AI Companion Robot** uses **PictoBlox emotion tracking** to monitor students daily, analyzes emotional patterns to detect long-term sadness, interacts friendly with students to improve mood using games and brain exercises, monitors heart rate, temperature, and stress, and sends **SMS alerts to parents and the principal** through IoT if the student needs help.

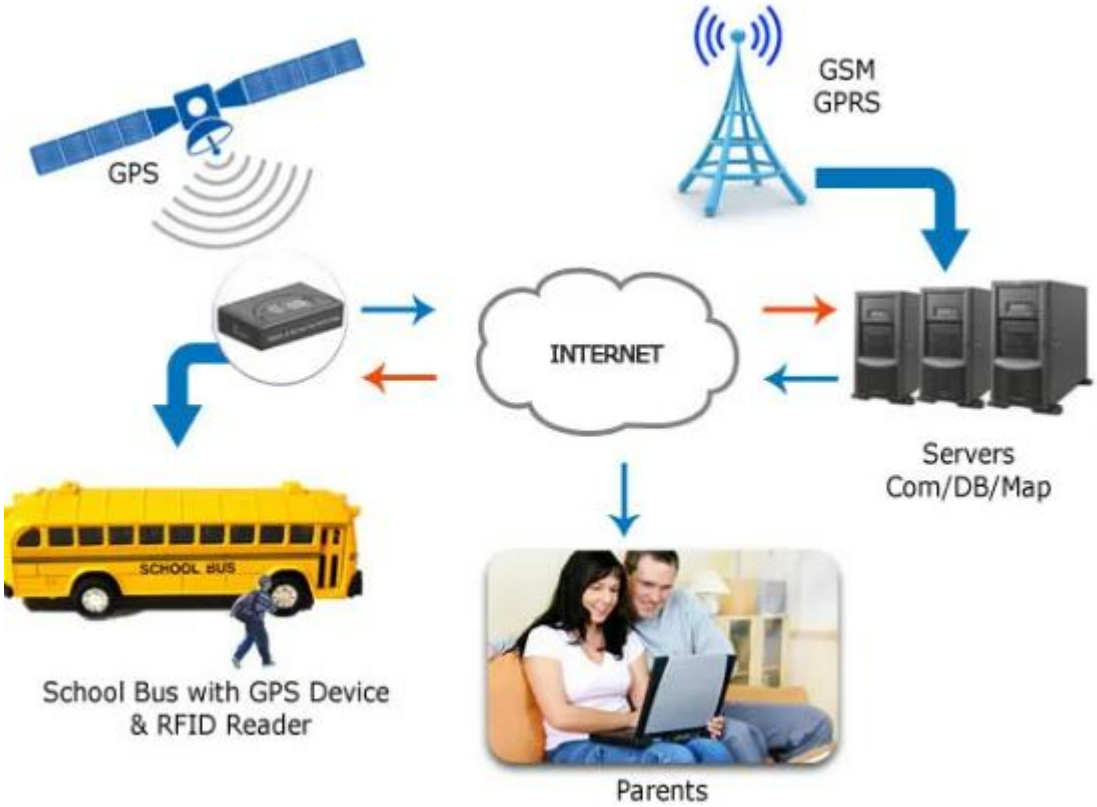


The system uses Arduino Uno and ESP32 (Wi-Fi & IoT) as controllers, a USB camera/webcam for emotion detection, heartbeat (pulse) sensor, GSR sensor, and temperature sensor (LM35/DS18B20) to monitor student health, a 16×2 I2C LCD, speaker/buzzer, and LED indicators for output and interaction, powered by a rechargeable battery pack with power switch and charging module, all connected using a breadboard and jumper wires, and housed inside a robot body/enclosure (optional).

Students & Women Safety GPS Tracking System

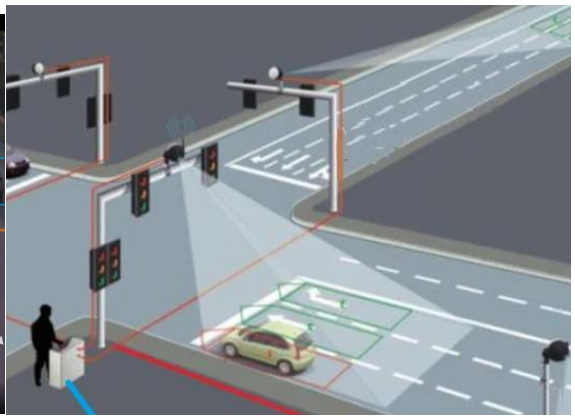
The Students and Women Safety GPS Tracking System continuously tracks live location using GPS, allows the user to send an emergency alert with one-button press, automatically shares location via SMS/IoT to parents, guardians, or police, and ensures quick response during danger situations.

- GPS module gets real-time location
- ESP32/Arduino processes data
- On panic button press, system sends SMS with Google Maps link
- Location can be tracked continuously through IoT
- Buzzer alerts nearby people for immediate help



The system uses Arduino Uno or ESP32 as the controller, a NEO-6M GPS module for live location tracking, a GSM module (SIM800L/SIM900) or ESP32 Wi-Fi for sending emergency alerts, a panic button for instant help activation, buzzer and LED for local alerts, powered by a rechargeable battery pack with power switch and charging module, all connected using a breadboard and jumper wires, with alerts received on a mobile phone, and housed in a wearable case or small enclosure.

Traffic Management System using PictoBlox ML (H4H)



The **AI-based Traffic Management System** uses **PictoBlox Machine Learning (H4H – Human for Human)** to analyze real-time traffic density through a camera, intelligently control traffic signals to reduce congestion, prioritize emergency vehicles, and improve road safety for humans.

- Camera captures live traffic images
- PictoBlox ML detects **vehicle count & density**
- AI decides **green signal time dynamically**
- Signals change automatically to reduce waiting time
- Emergency vehicles get priority